Year 1 AS Pure Maths

Chapter 4

Graphs and transformations

Practice Booklet

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4.1 Cubic graphs

Practice Book, printed page 26–27.

1. Sketch the following curves and indicate clearly any points of intersection with the axes.

a. $y = (x + 1)(x - 2)(x + 3)$

b. $y = x(2x - 1)(2x + 1)$

c. $y = (x - 2)(x + 4)^2$

Hint

Set each factor equal to 0 and solve to find the x-coordinates of the points where the graph crosses the x-axis. Substitute $x = 0$ into the equation to find the y-intercept.

2. Factorise each equation fully and sketch the curve:

a. $y = x^3 + 2x^2 - 3x$

b. $y = -x^3 - 2x^2 + 3x$

c. $y = 6x^3 - 3x^2$

Hint

One of the factors will be x, so the curve will pass through the origin.

3. Sketch the following curves and indicate clearly any points of intersection with the axes.

a. $y = (x - 2)(x^2 + x + 3)$

b. $y = (x - 3)^3$

c. $y = -(x - 4)^3$

Hint

The curve given by the equation $y = (x - a)^3$ is obtained when the graph of $y = x^3$ is translated a units to the right.

4. E

a. Factorise completely $x^3 + 8x^2 + 16x$. (2 marks)

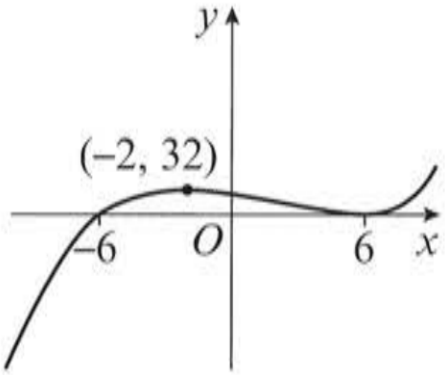
b. Sketch the curve with equation $y = x^3 + 8x^2 + 16x$, showing the coordinates of the points at which the curve cuts or touches the x-axis. (2 marks)

5. E

a. Factorise completely $25x - 4x^3$. (3 marks)

b. Sketch the curve C with equation $y = 25x - 4x^3$.
Show on your sketch the coordinates at which the curve meets the x-axis. (3 marks)

6. E/P The diagram shows a graph with equation $y = ax^3 + bx^2 + cx + d$, where a, b, c and d are real constants.



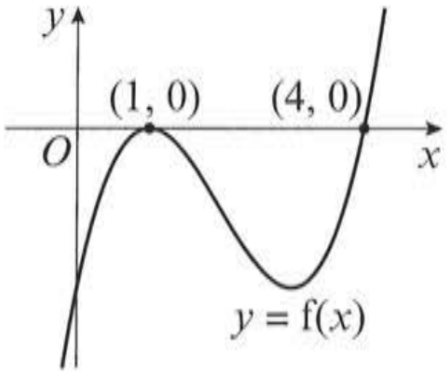
The graph passes through the points $(-6, 0)$ and $(-2, 32)$ and touches the x-axis at the point $(6, 0)$.
A student attempts to find the equation of the curve, and writes the following working:

$$y = (x + 6)(x - 6)^2$$
$$y = (x + 6)(x^2 - 12x + 36)$$
$$y = x^3 - 6x^2 - 36x + 216$$

a. Explain the mistake the student has made. (1 mark)

b. Find the correct equation of the curve. (4 marks)

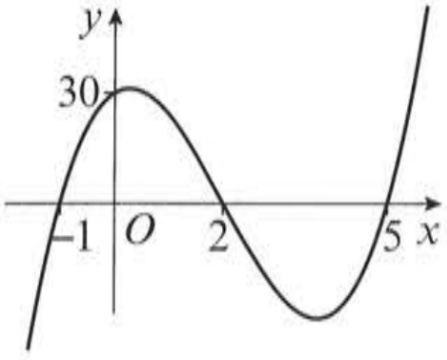
7. E/P The diagram shows a sketch of the curve with equation $y = f(x)$.
The curve touches the x-axis at the point $(1, 0)$ and passes through the point $(4, 0)$.



a. The equation of the curve can be written in the form $y = x^3 - ax^2 + bx - c$, where a, b and c are positive integers.
Calculate the values of a, b and c. (4 marks)

b. Write down the coordinates of the point where the graph crosses the y-axis. (1 mark)

8. E/P The diagram shows the graph of $y = ax^3 - bx^2 + cx + d$, where a, b, c and d are real constants.
Find the values of a, b, c and d. (4 marks)



4.2 Quartic graphs

Practice Book, printed page 27–28.

1. Sketch the following curves and indicate clearly any points of intersection with the axes.

a. $y = (x + 1)(x + 3)(x - 1)(x - 3)$

b. $y = -(x - 2)^2(x - 5)^2$

c. $y = (2x + 1)^3(x - 3)$

Hint

A quartic graph can be written in the form $y = ax^4 + bx^3 + cx^2 + dx + e$, where a, b, c, d and e are real constants and $a \neq 0$. If the right-hand side of the equation is factorised, you can determine any points of intersection with the x-axis by setting each factor equal to 0.

2. Sketch the curves and indicate clearly any points of intersection with the axes.

a. $y = (x^2 - x - 2)(x^2 + x - 12)$

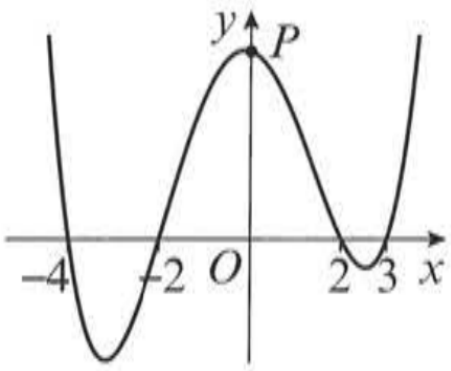
b. $y = (x - 5)^2(x^2 - 3x + 2)$

c. $y = (2x + 5)^4$

Hint

Fully factorise the equations first. In part **c**, the curve only touches the x-axis at one point.

3. The graph of $y = x^4 + x^3 - cx^2 - dx + e$ is shown in the diagram, where c, d and e are real constants. The curve intersects the y-axis at the point P.



a. Find the coordinates of point P.

b. Find the values of c, d and e.

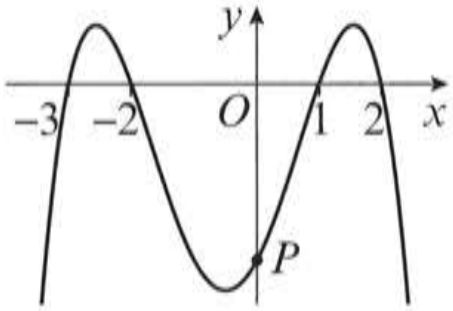
Hint

Write the equation as a product of its factors, then multiply out the brackets to find the coefficients of each term.

4. E/P Sketch the graph of $y = (x + 3)(x - 4)(x^2 + 7x + 10)$. (3 marks)

5. E/P Sketch the graph of $y = -(x + 2)^2(x^2 - 4x + 3)$. (3 marks)

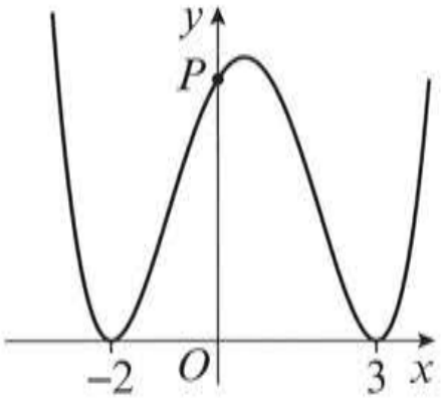
6. E/P The graph of $y = -x^4 - bx^3 + cx^2 + dx - e$ is shown in the diagram, where b, c, d and e are real constants. The curve intersects the y-axis at the point P.



a. Find the coordinates of P. (2 marks)

b. Find the values of b, c, d and e. (3 marks)

7. E/P The graph of $y = x^4 - bx^3 - cx^2 + dx + e$ is shown in the diagram, where b, c, d and e are real constants. The curve intersects the y-axis at the point P.



a. Find the coordinates of P. (2 marks)

b. Find the values of b, c, d and e. (3 marks)

4.3 Reciprocal graphs

Practice Book, printed page 28.

1. Sketch on the same diagram:

a. $y = \frac{2}{x}$

b. $y = \frac{6}{x}$

Hint

These are graphs of the type $y = \frac{k}{x}$ with $k > 0$. The curves will appear in the first and third quadrants, and the greater the value of k, the further the curve will be from the origin.

2. Sketch on the same diagram:

a. $y = -\frac{3}{x}$

b. $y = -\frac{7}{x}$

Hint

These are graphs of the type $y = \frac{k}{x}$ with $k < 0$. The curves will appear in the second and fourth quadrants, and the smaller the value of k, the further the curve will be from the origin.

3. Sketch on the same diagram:

a. $y = \frac{5}{x^2}$

b. $y = \frac{10}{x^2}$

Hint

These are graphs of the type $y = \frac{k}{x^2}$ with $k > 0$. The y-values are always positive, so the curves will appear in the first and second quadrants.

4. Sketch on the same diagram:

a. $y = -\frac{1}{x^2}$

b. $y = -\frac{9}{x^2}$

Hint

These are graphs of the type $y = \frac{k}{x^2}$ with $k < 0$. The y-values are always negative, so the curves will appear in the third and fourth quadrants.

5. E

a. On the same set of axes, sketch the graphs of $y = \frac{1}{x}$ and $y = \frac{4}{x^2}$ (4 marks)

b. State the equations of the asymptotes. (2 marks)

6. E

a. On the same set of axes, sketch the graphs of $y = -\frac{3}{x}$ and $y = -\frac{15}{x^2}$ (4 marks)

b. State the equations of the asymptotes. (2 marks)

4.4 Points of intersection

Practice Book, printed page 29–30.

1. In each case:

i sketch the two curves on the same axes

ii find the coordinates of the points of intersection.

a. $y = x(x - 4)$ and $y = x^2(3 - x)$

b. $y = x^2 + x - 2$ and $y = x^3 + x^2 - 2x$

Hint

If the coordinates of the points of intersection are not clear from your sketch, set the equations equal and solve to find the x-coordinates. Then substitute these values to find the y-coordinates.

2.

a. On the same set of axes, sketch the curves with equations $y = x^3$ and $y = \frac{5}{x^2}$

b. Show that the x-coordinates at the points of intersection of these curves satisfy the equation $x^5 - 5 = 0$.

c. State the number of real solutions to this equation.

Hint

For part **b**, set the equations equal to each other and rearrange to equal 0.

3.

a. On the same set of axes, sketch the curves with equations $y = -\frac{3}{x^2}$ and $y = -x^2(2x + 5)$

b. Show that the x-coordinates at the points of intersection of these curves satisfy the equation $x^4(2x + 5) - 3 = 0$.

c. State the number of real solutions to this equation.

Hint

The number of real solutions is the same as the number of times the two curves intersect.

4. In each case:
i sketch the curves with equations $y = f(x)$ and $y = g(x)$ on the same set of axes
ii state, giving a reason, the number of real solutions to the equation $f(x) = g(x)$.

a. $f(x) = -\frac{1}{x}$ and $g(x) = (x+1)(x-3)^2$

b. $f(x) = \frac{7}{x^2}$ and $g(x) = x^2(4x-9)$

Hint

The number of points of intersection of the curves $y = f(x)$ and $y = g(x)$ will be the same as the number of real solutions to the equation $f(x) = g(x)$.

5. E/P The curve C has equation $y = \frac{2}{x} - 5$, $x \neq 0$, and the line l has equation $y = 2x - 5$.
Find the coordinates of the points of intersection of $y = \frac{2}{x} - 5$ and $y = 2x - 5$. (5 marks)

6. E/P

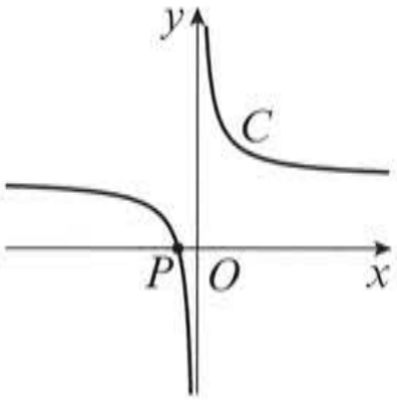
a. On the same set of axes, sketch the graphs of $y = x^2(x^2 - x - 2)$ and $y = -\frac{1}{4}x^3$ (5 marks)

b. Write down the number of real solutions to the equation $-4x^2(x^2 - x - 2) = x^3$ (1 mark)

c. Show that the equation $-4x^2(x^2 - x - 2) = x^3$ can be rearranged to give $x^2(4x^2 - 3x - 8) = 0$ (2 marks)

d. Hence determine the x-coordinates of the points of intersection, giving your answers exactly. (3 marks)

7. E/P The diagram shows a sketch of the curve C with equation $y = \frac{1}{x} + 2$, $x \neq 0$.
The curve crosses the x-axis at the point P.



a. **i** State the x-coordinate of the point P.
ii State the equations of the asymptotes of the curve. (2 marks)

The curve D has equation $y = x^2 + 2$ for all real values of x.

b. On the same set of axes, copy the graph of curve C and sketch the graph of curve D.
Mark clearly where curve D crosses the y-axis. (3 marks)

c. Find the coordinates of the point(s) of intersection of the graphs. (2 marks)

8. E/P

a. On the same set of axes, sketch the graphs of:
i $y = x(x+3)(2-x)$ **ii** $y = -\frac{4}{x}$
In each case, show clearly the coordinates of all the points where the curves cross the coordinate axes. (6 marks)

b. Using your sketch, state, giving a reason, the number of real solutions to the equation $x(x+3)(2-x) + \frac{4}{x} = 0$ (2 marks)

9. E/P

a. Sketch and clearly label the graphs of $y = \frac{2}{x}$ and $y = 4x - 2$ on a single diagram. (4 marks)

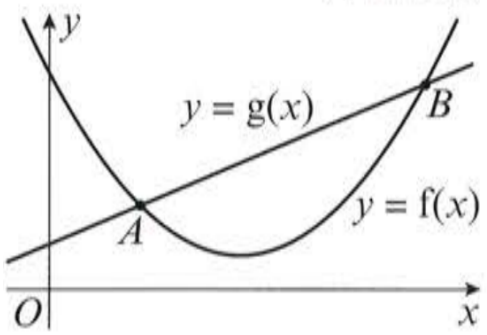
b. Find the coordinates of the points of intersection of $y = \frac{2}{x}$ and $y = 4x - 2$. (4 marks)

10. E/P Find the coordinates of any points of intersection between the curve with equation $y = x^2 - 10x - 15$ and the straight line with equation $y = 10 - 10x$. (4 marks)

11. E/P Two functions $f(x)$ and $g(x)$ are given by $f(x) = x^2 - 9x + 24$ and $g(x) = 2x + a$, where a is a real constant.

a. Given that the graphs of $y = f(x)$ and $y = g(x)$ intersect at two distinct points, find the range of possible values of a. (4 marks)

The diagram shows the curves $y = f(x)$ and $y = g(x)$ in the case where $a = 5$.



b. Using an appropriate algebraic method, find the coordinates of points A and B. Give your answers correct to 3 significant figures. (3 marks)

4.5 Translating graphs

Practice Book, printed page 30–32.

1. Given that:

a. $f(x) = x^2$

b. $f(x) = x^3$

c. $f(x) = \frac{1}{x}$

sketch each of the following graphs, stating any points where the curve crosses or touches the coordinate axes, and the equations of any asymptotes.

i $y = f(x) + 1$ **ii** $y = f(x - 2)$ **iii** $y = f(x + 3)$ **iv** $y = f(x) - 4$

Hint

The graph of $y = f(x) + a$ is a translation of the graph $y = f(x)$ by the vector $\begin{pmatrix} 0 \\ a \end{pmatrix}$.
The graph of $y = f(x + a)$ is a translation of the graph $y = f(x)$ by the vector $\begin{pmatrix} -a \\ 0 \end{pmatrix}$.

2. $f(x) = (x + 3)(x + 1)(x - 2)$, $g(x) = x(x + 3)$ and $h(x) = \frac{1}{x}$

Sketch the following graphs, labelling any points where the curve crosses the axes, and stating the equations of any asymptotes.

a. $y = f(x - 3)$

b. $y = g(x + \frac{1}{2})$

c. $y = h(x) - 5$

Hint

Asymptotes will be translated by the same vector as the curve.

3.

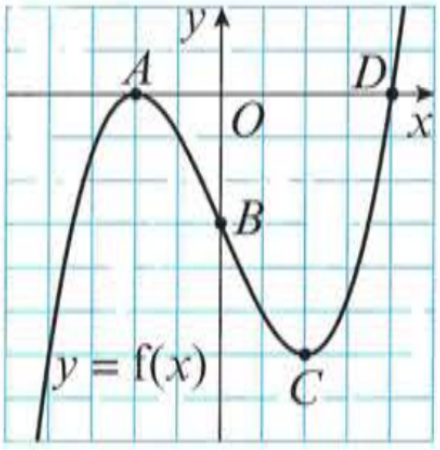
a. Sketch the graph of $y = f(x)$ where $f(x) = x^2(x - 2)(x + 3)$

b. Sketch the curve with equation $y = f(x - 2)$

Hint

Write the equation of the transformed function and substitute $x = 0$ into the equation to find the y-intercept of the curve.

4. E/P It is given that $f(x) = 3x^3 - 9x - 6$.
The diagram shows a sketch of part of the curve with equation $y = f(x)$.

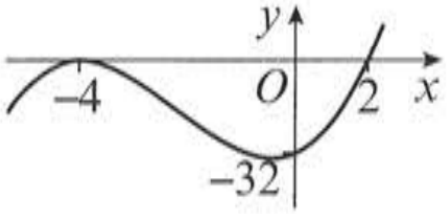


On a separate set of axes, sketch the following curves showing the new coordinates of the images of points A, B, C and D.

a. $y = f(x) + 4$ (2 marks)

b. $y = f(x - 3)$ (2 marks)

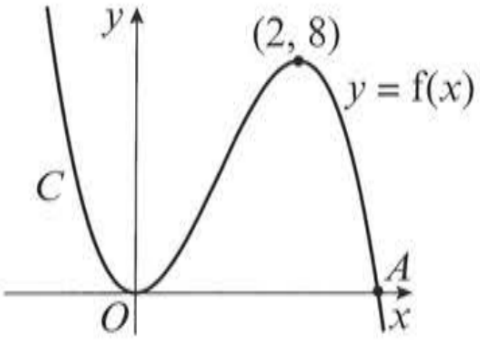
5. E/P The diagram shows a sketch of the curve with equation $y = f(x)$.
The curve crosses the x-axis at $(2, 0)$, touches it at $(-4, 0)$ and crosses the y-axis at $(0, -32)$.



a. Write down the equation of the curve in the form $y = (x + a)^2(x - b)$ where a and b are integers. (2 marks)

b. Sketch the curve with equation $y = f(x - 2)$, clearly showing the coordinates of the points where the curve intersects with the axes. (4 marks)

6. E/P The diagram shows a sketch of the curve C with equation $y = f(x)$, where $f(x) = x^2(6 - 2x)$.
There is a local minimum at the origin, a local maximum at the point $(2, 8)$ and C cuts the x-axis at the point A.

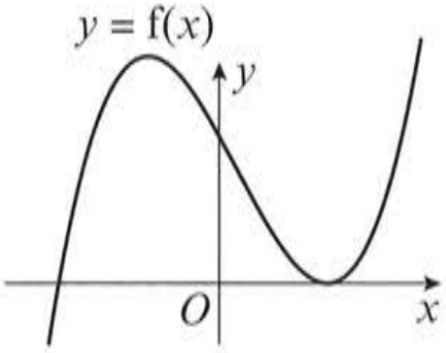


a. Write down the coordinates of the point A. (1 mark)

The curve with equation $y = f(x) - k$, where k is a constant, has a local maximum at $(2, 1)$.

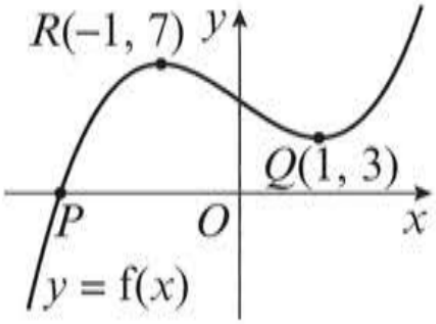
b. Write down the value of k. (1 mark)

7. E/P The diagram shows the curve with equation $y = f(x)$, where $f(x) = (x+3)(x-2)^2$



Given that the graph of $y = f(x+a)$ passes through the origin, find the possible values of a. (2 marks)

8. E/P The sketch shows a cubic curve with equation $y = f(x)$.
The curve crosses the x-axis at P and has turning points at $Q(1,3)$ and $R(-1,7)$.



Given that the equation $f(x) + a = 0$, where a is a real constant, has three distinct real roots, write down the range of possible values of a. (2 marks)

4.6 Stretching graphs

Practice Book, printed page 32–33.

1. Given that:

a. $f(x) = (x-1)^2$

b. $f(x) = x^3 + 1$

c. $f(x) = \frac{1}{x-1}$

sketch each of the following graphs, stating any points where the curve crosses or touches the coordinate axes, and the equations of any asymptotes.

i $y = 3f(x)$ **ii** $y = f(\frac{1}{2}x)$ **iii** $y = -f(x)$ **iv** $y = f(-x)$

Hint

The graph of $y = af(x)$ is a stretch of the graph $y = f(x)$ by a scale factor of a in the vertical direction.
The graph of $y = f(ax)$ is a stretch of the graph $y = f(x)$ by a scale factor of $\frac{1}{a}$ in the horizontal direction.

2. $f(x) = x^2 - 16$

a. Sketch the curve $y = f(x)$.

b. Sketch these graphs and find the equations of the curves in terms of x.
i $y = f(\frac{1}{2}x)$ **ii** $y = \frac{1}{4}f(x)$ **iii** $y = -f(x)$

c. Use the equations found in part **b** to find the coordinates of the y-intercept of each curve.

Hint

First find the equation of the transformed function.
Substitute $x = 0$ to find the point of intersection of the curve with the y-axis.

3.

a. Sketch the curve C with equation $y = x^2(x - 5)$.

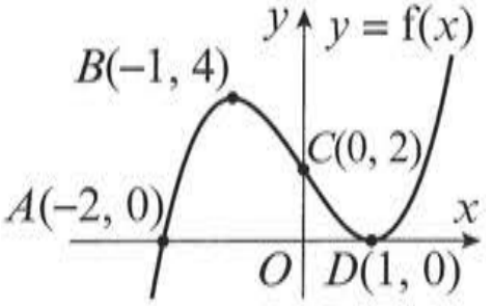
b. On the same set of axes, sketch the curve D with equation $y = (4x)^2(4x - 5)$.

c. Describe the transformation from curve C to curve D.

Hint

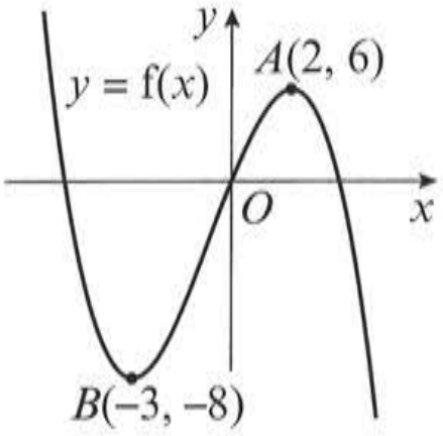
To write the transformation in terms of $f(x)$, determine what x has been multiplied by to obtain the transformed equation.

4. E The diagram shows a sketch of part of the curve with equation $y = f(x)$.



On a separate set of axes, sketch the curve with equation $y = f(\frac{1}{2}x)$, showing the location and coordinates of the images of points A, B, C and D. (2 marks)

5. E The diagram shows a sketch of part of the curve with equation $y = f(x)$.
The curve has a local maximum at $A(2, 6)$, a local minimum at $B(-3, -8)$ and passes through the origin O. On separate diagrams, sketch the curves with equations:

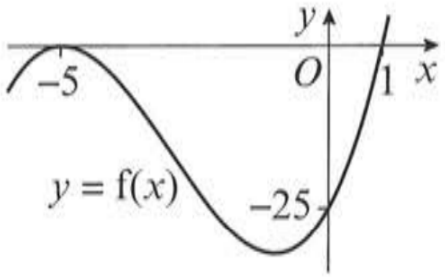


a. $y = f(\frac{1}{2}x)$ (2 marks)

b. $y = f(-x)$ (2 marks)

On each diagram, show clearly the locations of the y-intercept, and any local maximum and local minimum points.

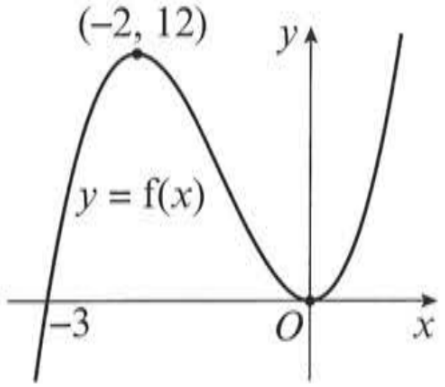
6. E The diagram shows the curve with equation $y = f(x)$ where $f(x) = (x+5)^2(x-1)$, $x \in \mathbb{R}$. The curve crosses the x-axis at $(1, 0)$, touches it at $(-5, 0)$ and crosses the y-axis at -25.



a. Sketch the curve C with equation $y = 3f(x)$ and state the coordinates of the points where the curve C meets the axes. (3 marks)

b. Write down an equation of the curve C. (1 mark)

7. E/P The diagram shows the graph of the curve $y = f(x)$.
The graph crosses the x-axis at $(-3, 0)$ and has turning points at $(-2, 12)$ and $(0, 0)$.



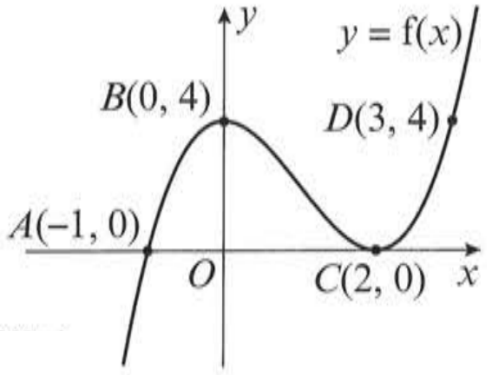
a. The graph of $y = f(ax)$ has a turning point at $(-\frac{1}{4}, 12)$.
Find the value of a. (1 mark)

b. The graph of $y = bf(x)$ has a turning point at $(-2, 6)$.
Find the value of b. (1 mark)

4.7 Transforming functions

Practice Book, printed page 33–35.

1. The diagram shows a sketch of the curve $y = f(x)$.
The points $A(-1, 0)$, $B(0, 4)$, $C(2, 0)$ and $D(3, 4)$ lie on the curve.
Sketch the following graphs, showing the new coordinates of A, B, C and D:

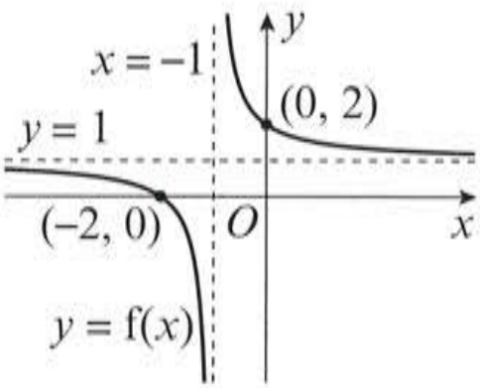


a. $y = f(x - 2)$
b. $y = f(x) + 3$
c. $y = f(2x)$

Hint

Translate or stretch the points A, B, C and D and then sketch the curve through these points.

2. The diagram shows the curve $y = f(x)$ which crosses the axes at $(-2, 0)$ and $(0, 2)$, and has asymptotes at $y = 1$ and $x = -1$.
Sketch these transformed graphs, showing the coordinates of intersections with the axes and the equations of any asymptotes:

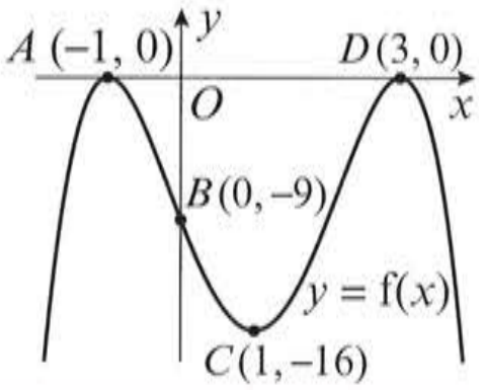


a. $y = f(-x)$
b. $y = 3f(x)$
c. $y = f(\frac{1}{2}x)$

Hint

Draw any transformed asymptotes first.

3. The diagram shows a sketch of the curve $y = f(x)$.
The points $A(-1, 0)$, $B(0, -9)$, $C(1, -16)$ and $D(3, 0)$ lie on the curve.
Give the coordinates of the points corresponding to A, B, C and D on each of the following transformed curves.

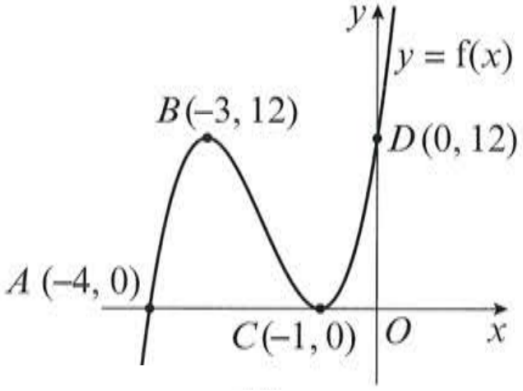


a. $2y = f(x)$
b. $\frac{1}{4}y = f(x)$
c. $y + 1 = f(x)$

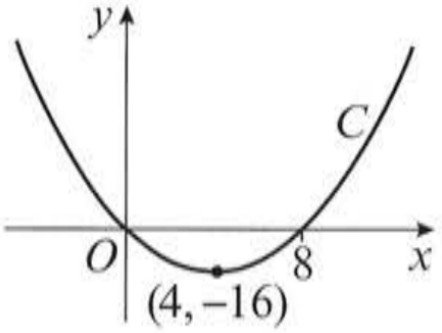
Hint

Rearrange the equations into the form $y = \ldots$

4. E/P The diagram shows a sketch of part of the curve with equation $y = f(x)$.
Sketch the curve with equation $y = f(x - 3)$, showing the location and coordinates of the images of points A, B, C and D. (4 marks)



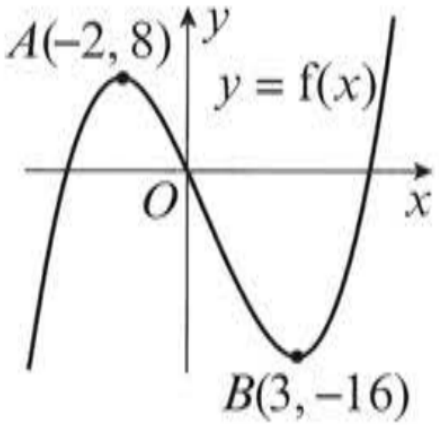
5. E/P The diagram shows a sketch of the curve C with equation $y = f(x)$.
The curve C passes through the origin and the point $(8, 0)$ and has a local minimum at the point $(4, -16)$. On separate diagrams, sketch the curves with equations:



a. $y = f(-x)$ (3 marks)
b. $y = f(x - a)$, where a is a constant and $a > 0$. (4 marks)

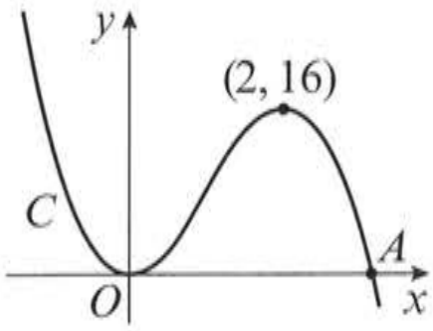
On each diagram, show the locations of any points where the curve intersects the x-axis and of any local minimum or local maximum points.

6. E/P The diagram shows a sketch of part of the curve with equation $y = f(x)$.
The curve has a local maximum at $A(-2, 8)$ and a local minimum at $B(3, -16)$ and passes through the origin O.



On a separate diagram, sketch the curve with equation $y = -f(2x)$, showing clearly the coordinates of the local maximum and the local minimum points and the location of the point where the curve crosses the y-axis. (4 marks)

7. E/P The diagram shows a sketch of the curve C with equation $y = f(x)$, where $f(x) = x^2(12 - 4x)$.
Curve C has a local minimum at the origin, a local maximum at the point $(2, 16)$ and cuts the x-axis at the point A.

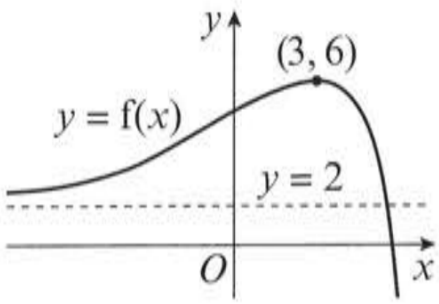


a. Write down the coordinates of the point A. (1 mark)

b. On separate diagrams sketch the curves with equations:
i $y = f(x - 2)$ **ii** $y = f(2x)$
On each sketch you should indicate clearly the location of the local maximum point and any points where the curve crosses or meets the coordinate axes. (6 marks)

c. The curve with equation $y = f(x) + k$, where k is a constant, has a local maximum point at $(2, 5)$. Write down the value of k. (1 mark)

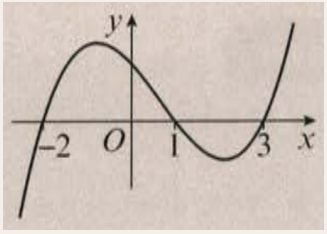
8. E/P The sketch shows part of the graph of the curve $y = f(x)$.
The curve has local maximum at $(3, 6)$ and asymptote $y = 2$.
The equation $f(x) + a = 0$, where a is a real constant, has exactly one real solution.
Find the range of possible values of a, giving your answer in set notation. (3 marks)



Problem solving: Set A

Bronze

The graph of $y = x^3 + bx^2 + cx + d$ is shown, where b, c and d are real constants.

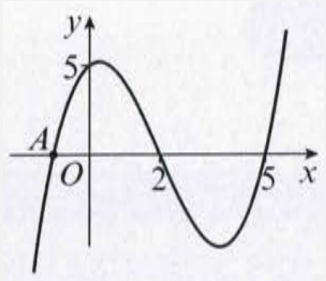


a. Find the values of b, c and d. (3 marks)

b. Hence find the coordinates of the point where the curve crosses the y-axis. (1 mark)

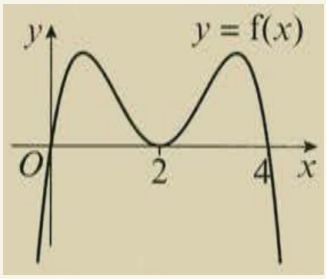
Silver

The graph of $y = x^3 + bx^2 + cx + d$ is shown, where b, c and d are real constants.
The curve crosses the x-axis at the points A, $(2, 0)$ and $(5, 0)$, and intersects the y-axis at $(0, 5)$.
Use the two known roots and the y-intercept to find the values of b, c and d. (4 marks)



Gold

The graph of $y = f(x)$ is shown, where $f(x) = ax^4 + bx^3 + cx^2 + dx$ and a, b, c and d are real constants.
The curve crosses the x-axis at $(0, 0)$ and $(4, 0)$ and touches the x-axis at $(2, 0)$.



a. Given that the equation of the curve can be written as $y = a(x^4 + mx^3 + nx^2 + px)$, where m, n, and p are real constants, find the values of m, n and p. (3 marks)

b. Given further that the equation $f(x) = 8$ has exactly two real solutions given by $x = 2 \pm \sqrt{2}$, find the value of a, and hence write down the equation of the curve. (2 marks)

Problem solving: Set B

Bronze

a. On the same set of axes, sketch the curves with equations $y = x^2(x - 2)$ and $y = 3x$. (3 marks)

b. Write down the number of points of intersection of these two curves, and hence the number of real solutions to the equation $x^2(x - 2) = 3x$. (1 mark)

c. Show that you can rearrange this equation to give $x(x + 1)(x - 3) = 0$. (2 marks)

d. Hence determine the exact coordinates of the points of intersection. (3 marks)

Silver

a. On the same set of axes, sketch the curves with equations $y = \frac{3}{(x-3)^2}$ and $y = \frac{6}{x}$ (4 marks)

b. Write down the number of real solutions to the equation $\frac{3x}{(x-3)^2} = 6$ (1 mark)

c. Determine the exact coordinates of the points of intersection of the two curves in part **a**. (4 marks)

Gold

The curve C has equation $y = \frac{4}{(x-2)^2}$. The line L has equation $y = 5 - x$.

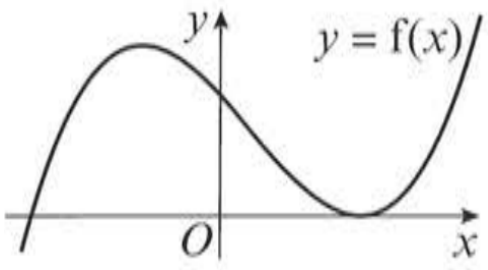
a. Given that the curve and the line intersect at the point $(1, 4)$, find the coordinates of any other points of intersection. (5 marks)

b. Hence sketch both graphs on the same set of axes, showing the coordinates of any points where the curve and line intersect each other, or either axis. (3 marks)

Selected exam question bank questions

Question 12

The diagram shows a sketch of part of the curve $y = f(x)$, $x \in \mathbb{R}$, where $f(x) = (2x - 3)^2(x + 2)$.

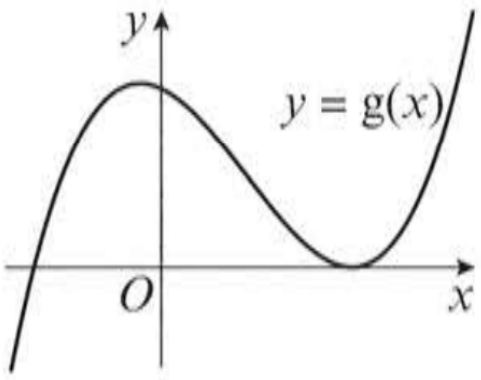


a. **i** Given that the curve with equation $y = f(x) - k$, $x \in \mathbb{R}$, passes through the origin, find the value of the constant k.
ii Given that the curve with equation $y = f(x + c)$, $x \in \mathbb{R}$, has a local minimum point at the origin, find the value of the constant c. (3 marks)

b. Show that $f'(x) = 12x^2 - 8x - 15$. (3 marks)

Question 22

The diagram shows a sketch of part of the curve $y = g(x)$ where $g(x) = (x + 1)(2x - 3)^2$



Deduce the values of x for which:

a. $g(x) \leqslant 0$ (1 mark)

b. $g(2x) = 0$ (3 marks)

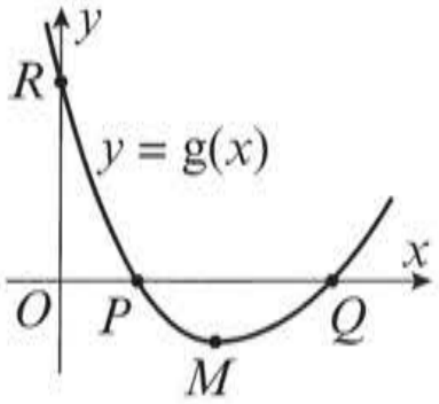
Question 33

The diagram shows a sketch of the curve with equation $y = g(x)$.

The curve has a single turning point, a local minimum, at the point $M(5, -1.5)$.

The curve crosses the x-axis at two points, $P(3, 0)$ and $Q(8, 0)$.

The curve crosses the y-axis at a single point $R(0, 9)$.



a. State the coordinates of the turning point on the curve with equation $y = 3g(x)$. (1 mark)

b. State the largest root of the equation $g(x - 2) = 0$. (1 mark)

c. State the range of values of x for which $g'(x) \geqslant 0$. (1 mark)

d. Given that the equation $g(x) + k = 0$, where k is a constant, has at least one real root, state the range of possible values of k. (1 mark)

Question 49

a. Factorise completely $36x - 25x^3$. (3 marks)

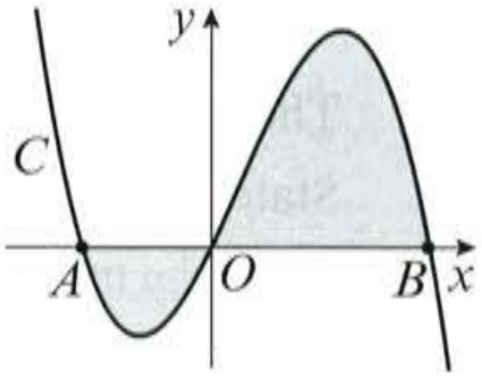
b. Sketch the curve C with equation $y = 36x - 25x^3$.
Show on your sketch the coordinates at which the curve meets the x-axis. (3 marks)

The points A and B lie on C and have x-coordinates of -1 and 1 respectively.

c. Show that the length of AB is $2\sqrt{k}$ where k is a constant to be found. (4 marks)

Question 72

The graph shows part of the curve C with equation $y = -x^3 + 2x^2 + 15x$.
The curve C crosses the x-axis at the origin O and at points A and B.



a. Using an appropriate algebraic method, find the coordinates of A and B. (3 marks)

b. The finite region shown shaded is bounded by the curve C and the x-axis.
Use calculus to find the total area of the shaded region. (8 marks)

Question 88

a. On separate sets of axes, sketch the graphs of:
i $y = -5x + c$, where c is a positive constant
ii $y = \frac{1}{x} + 6$
On each sketch, show the coordinates of any point at which the graph crosses the y-axis and the equation of any horizontal asymptote. (4 marks)

Given that $y = -5x + c$ meets the curve $y = \frac{1}{x} + 6$ at two distinct points,

b. show that $(6 - c)^2 > 20$ (3 marks)

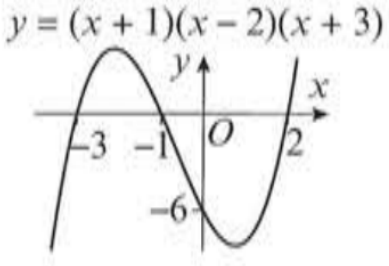
c. hence find the range of possible values for c. (4 marks)

Answers

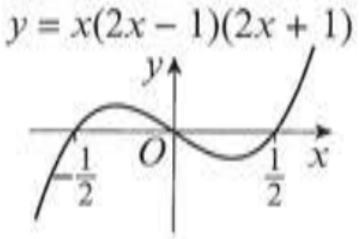
4.1 Cubic graphs

1.

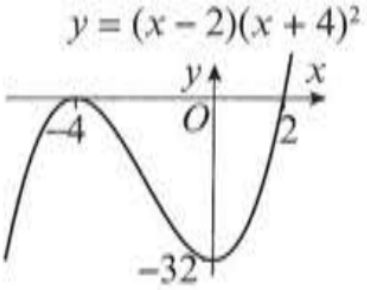
a.



b.

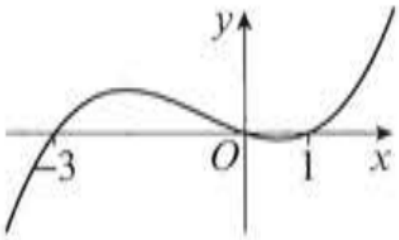


c.

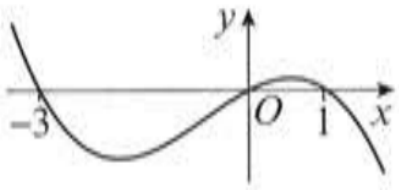


2.

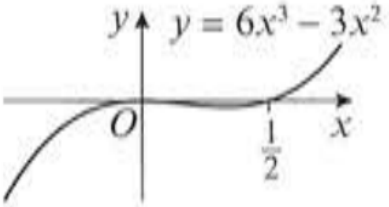
a. $y = x(x-1)(x+3)$



b. $y = x(1-x)(x+3)$

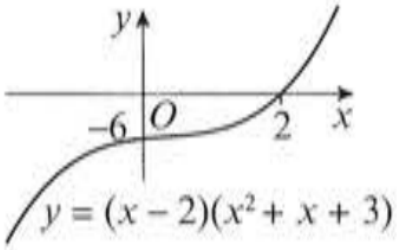


c. $y = 3x^2(2x-1)$

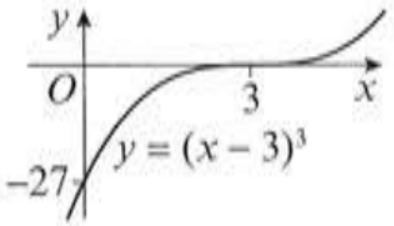


3.

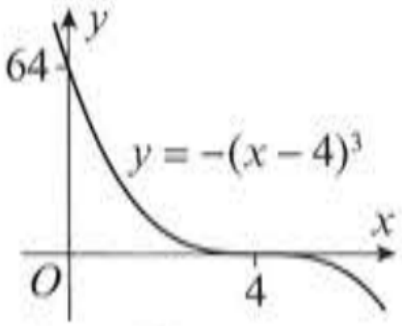
a.



b.



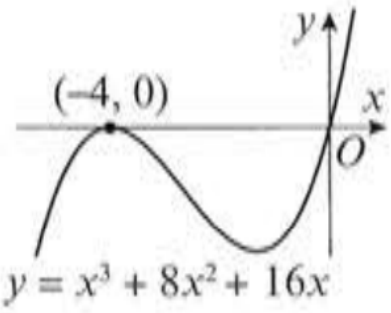
c.



4.

a. $x(x+4)^2$

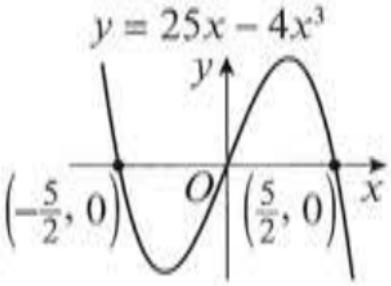
b.



5.

a. $x(5-2x)(5+2x)$

b.



6.

a. Assumed that the coefficient of x^3 is 1 when it isn't resulting in an equation which does not pass through $(-2, 32)$.

b. $y = \frac{1}{8}x^3 - \frac{3}{4}x^2 - \frac{9}{2}x + 27$

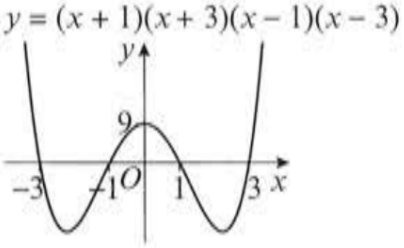
7. **a** $a = 6$, $b = 9$, $c = 4$ **b** $(0, -4)$

8. $a = 3$, $b = 18$, $c = 9$, $d = 30$

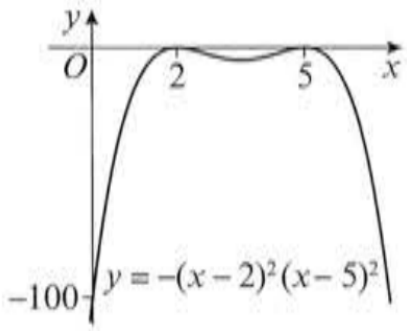
4.2 Quartic graphs

1.

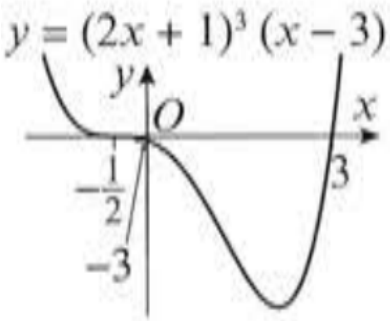
a.



b.

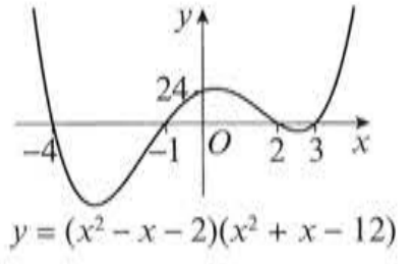


c.

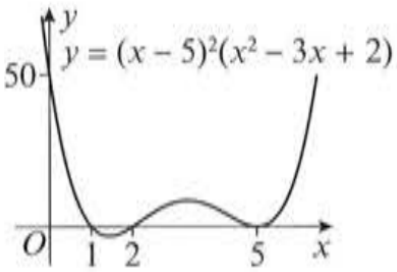


2.

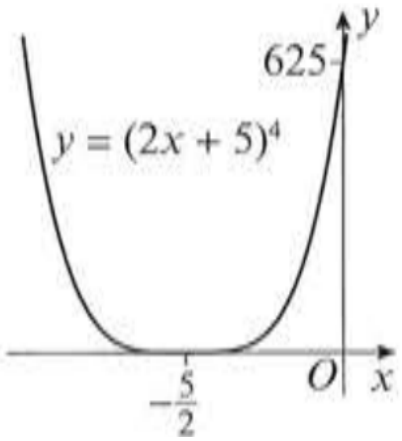
a.



b.

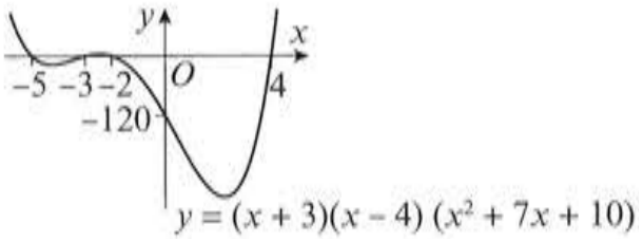


c.

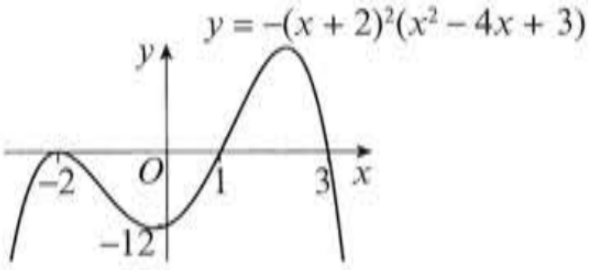


3. **a** $(0, 48)$ **b** $c = 16$, $d = 4$, $e = 48$

4.



5.

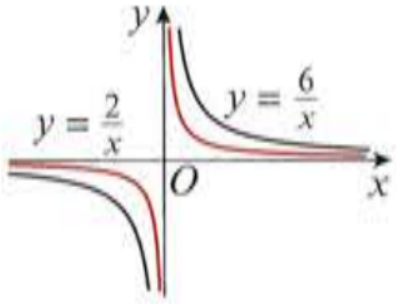


6. **a** $(0, -12)$ **b** $b = 2$, $c = 7$, $d = 8$, $e = 12$

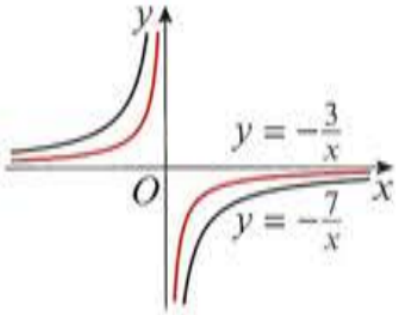
7. **a** $(0, 36)$ **b** $b = 2$, $c = 11$, $d = 12$, $e = 36$

4.3 Reciprocal graphs

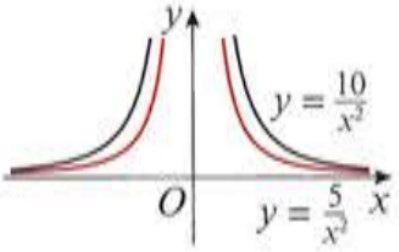
1.



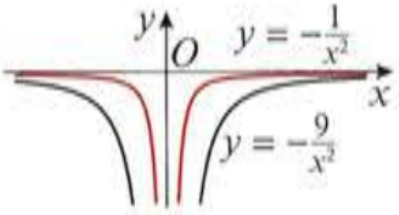
2.



3.

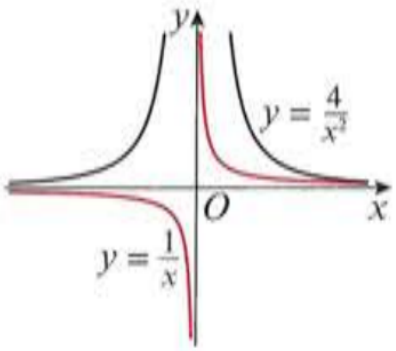


4.



5.

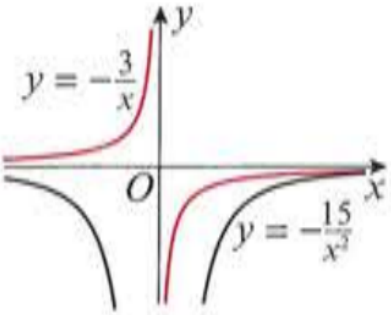
a.



b. Both graphs have asymptotes $y = 0$ and $x = 0$.

6.

a.

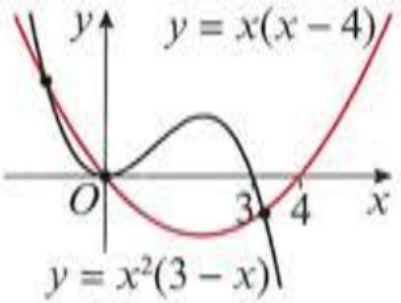


b. Both graphs have asymptotes $y = 0$ and $x = 0$.

4.4 Points of intersection

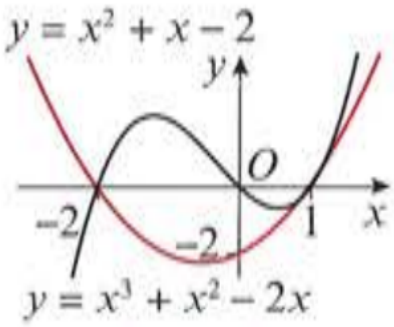
1.

a. i



ii $(0, 0)$, $(1 + \sqrt{5}, 2 - 2\sqrt{5})$ and $(1 - \sqrt{5}, 2 + 2\sqrt{5})$

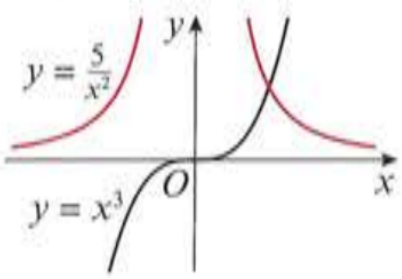
b. i



ii $(-2, 0)$, $(1, 0)$

2.

a.

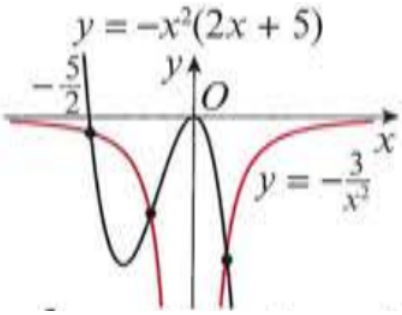


b. $x^3 = \frac{5}{x^2} \Rightarrow x^5 - 5 = 0$

c. 1

3.

a.

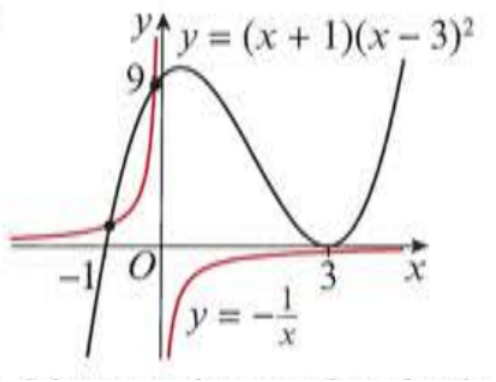


b. $-\frac{3}{x^2} = -x^2(2x + 5) \Rightarrow x^4(2x + 5) - 3 = 0$

c. 3

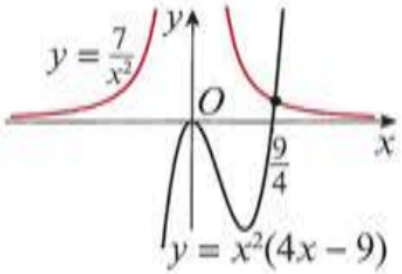
4.

a. i



ii 2 intersections ⇒ 2 real solutions

b. i

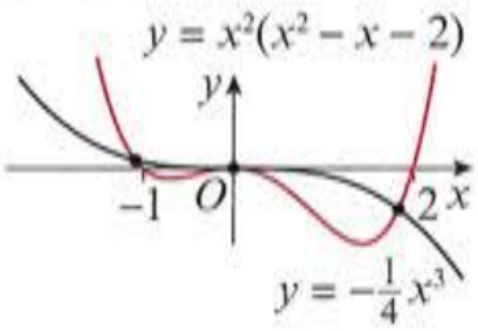


ii 1 intersection ⇒ 1 real solution

5. $(-1, -7)$, $(1, -3)$

6.

a.



b. 3

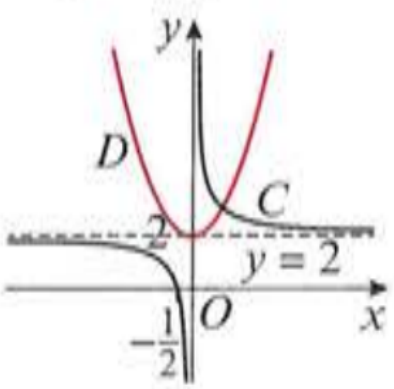
c. $-4x^2(x^2 - x - 2) = x^3 \Rightarrow x^3 + 4x^2(x^2 - x - 2) = 0$
$\Rightarrow 4x^4 - 3x^3 - 8x^2 = 0 \Rightarrow x^2(4x^2 - 3x - 8) = 0$

d. $x = 0,\ x = \frac{3 \pm \sqrt{137}}{8}$

7.

a. **i** $(-0.5, 0)$ **ii** $x = 0,\ y = 2$

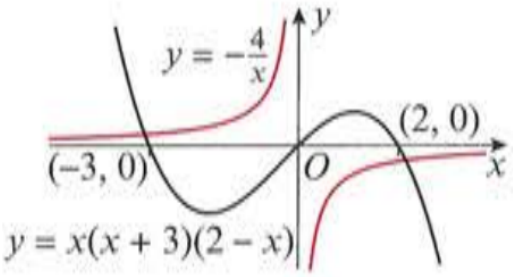
b.



c. $(1, 3)$

8.

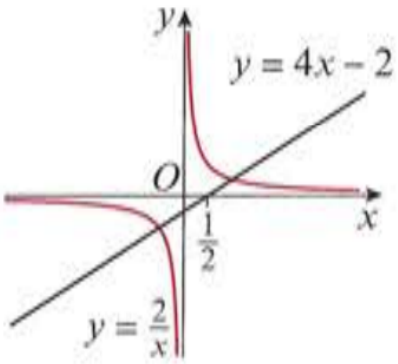
a.



b. 2 intersections $\Rightarrow$ 2 solutions

9.

a.



b. $(-0.5, -4)$ and $(1, 2)$

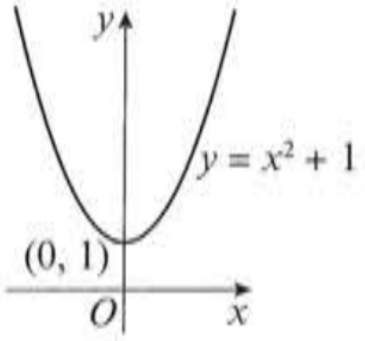
10. $(-5, 60)$ and $(5, -40)$

11. **a** $a > -6.25$ **b** $(2.15, 9.29)$ and $(8.85, 22.7)$

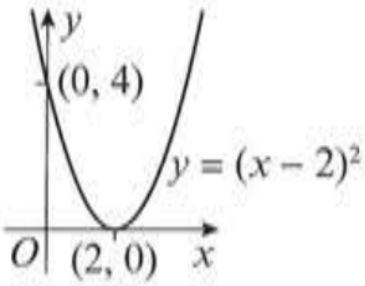
4.5 Translating graphs

1.

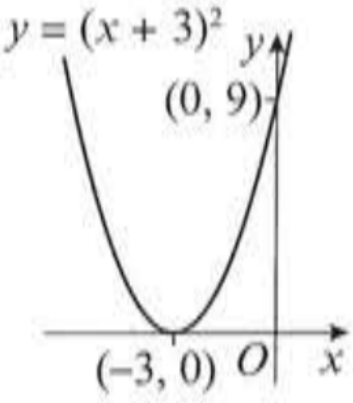
a. i



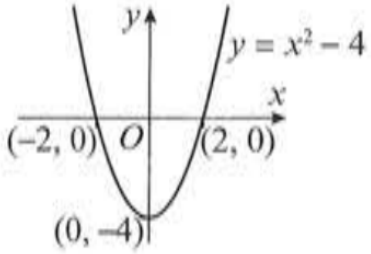
ii



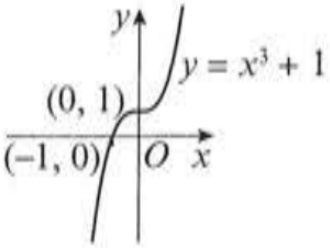
iii



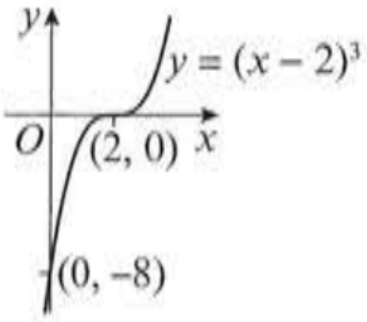
iv



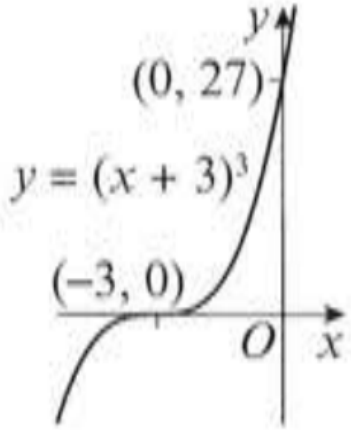
b. i



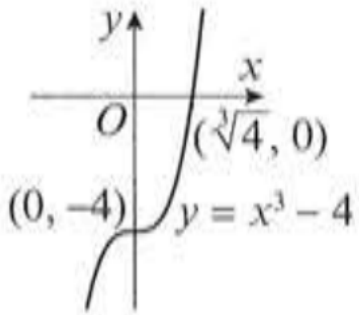
ii



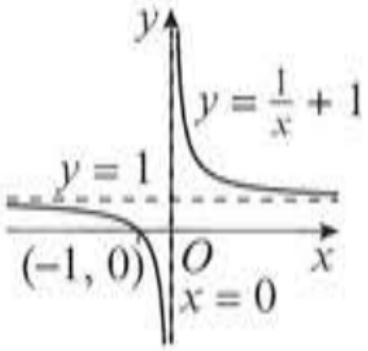
iii



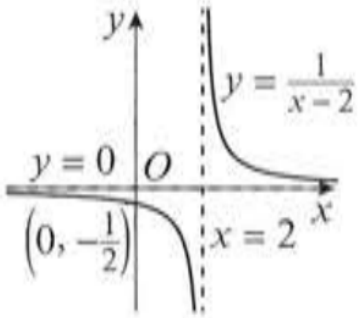
iv


y
x
O
$(\sqrt[3]{4}, 0)$
$(0, -4)$
$y = x^3 - 4$

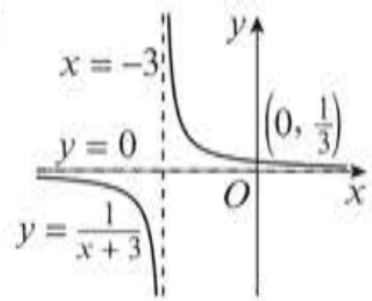
c. i


y
$y = \frac{1}{x} + 1$
$y = 1$
$(-1, 0)$
O
x
$x = 0$

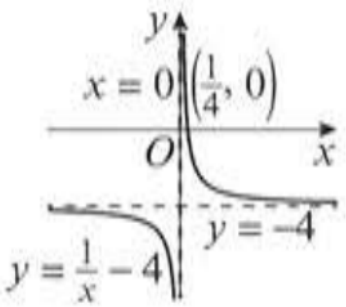
ii


y
$y = \frac{1}{x-2}$
$y = 0$
O
x
$\left(0, -\frac{1}{2}\right)$
$x = 2$

iii

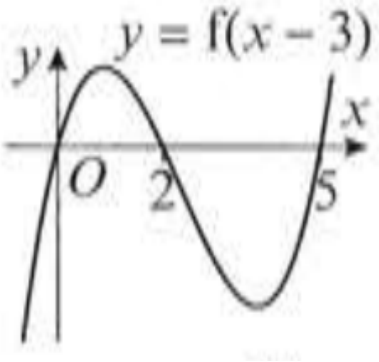

y
$x = -3$
$\left(0, \frac{1}{3}\right)$
$y = 0$
O
x
$y = \frac{1}{x+3}$

iv

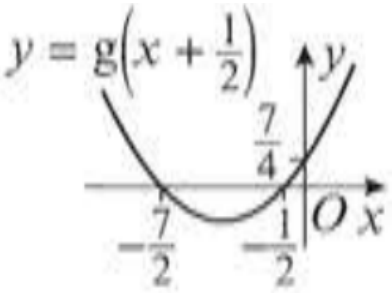


2.

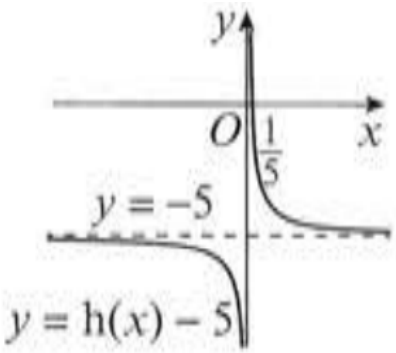
a.



b.

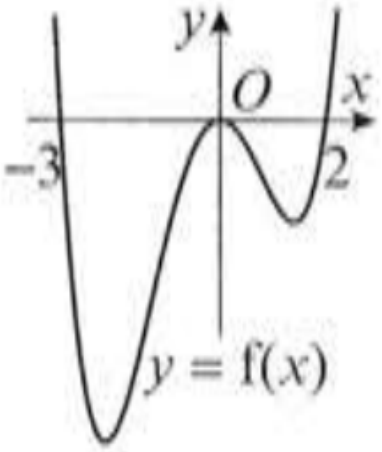


c.

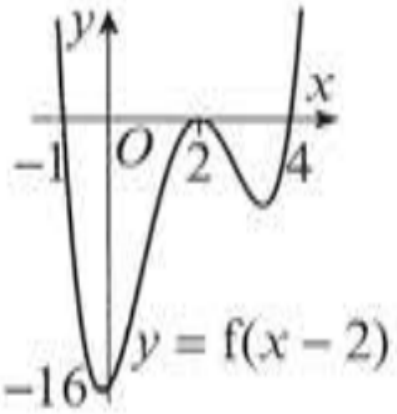


3.

a.

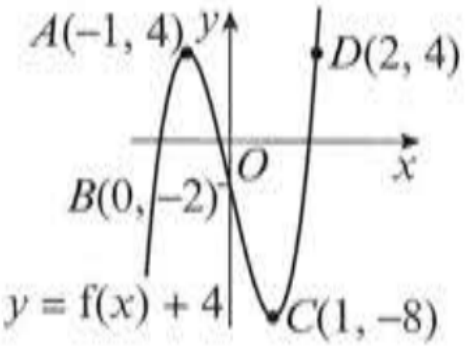


b.

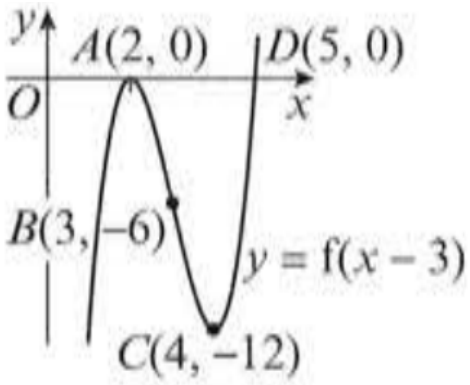


4.

a.



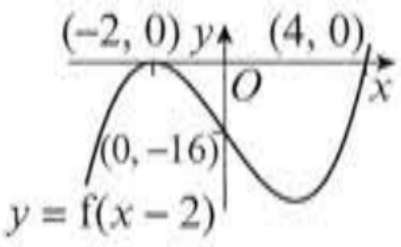
b.



5.

a. $y = (x + 4)^2(x - 2)$

b.



6. **a** $(3, 0)$ **b** 7

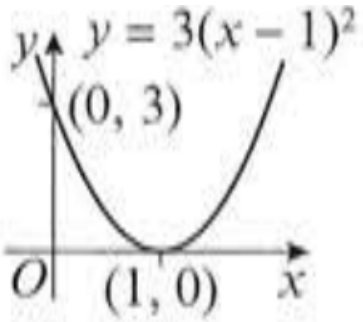
7. 2 and -3

8. $-7 < a < -3$

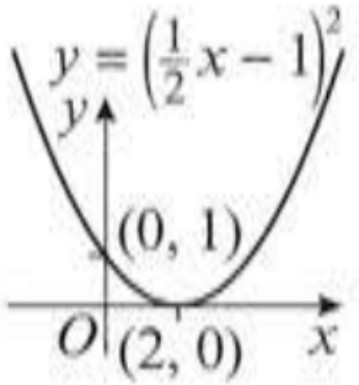
4.6 Stretching graphs

1.

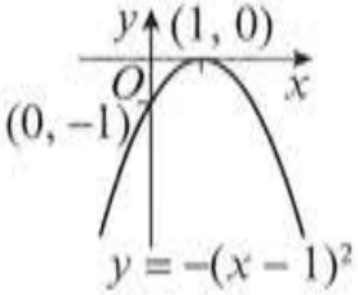
a. i



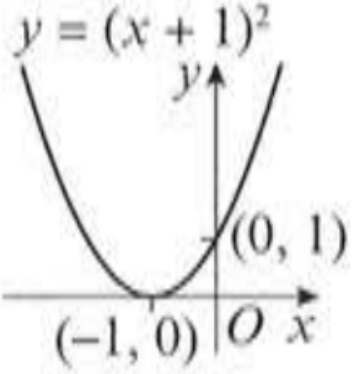
ii


$y = \left(\frac{1}{2}x - 1\right)^2$
y
(0, 1)
O
(2, 0)
x

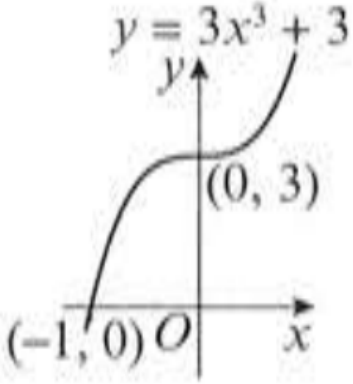
iii


y
(1, 0)
O
x
(0, −1)
$y = -(x - 1)^2$

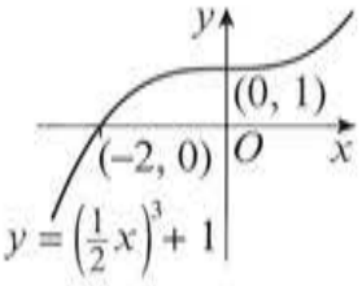
iv


$y = (x + 1)^2$
y
(0, 1)
(−1, 0)
O
x

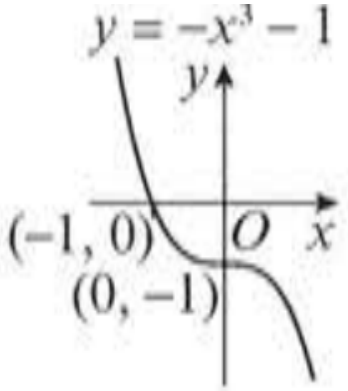
b. i


$y = 3x^3 + 3$
y
(0, 3)
(−1, 0)
O
x

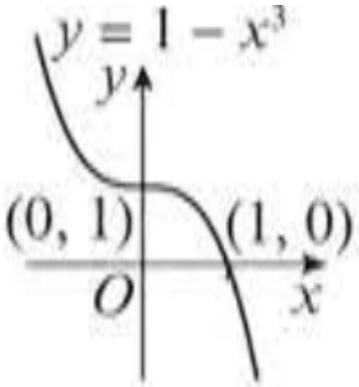
ii


y
(0, 1)
(−2, 0)
O
x
$y = \left(\frac{1}{2}x\right)^3 + 1$

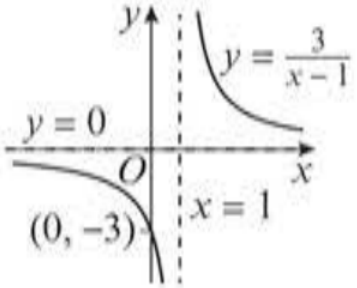
iii


$y = -x^3 - 1$
y
(−1, 0)
O
x
(0, −1)

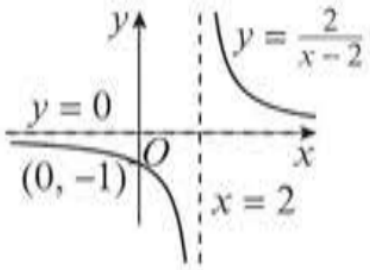
iv


$y = 1 - x^3$
y
(0, 1)
(1, 0)
O
x

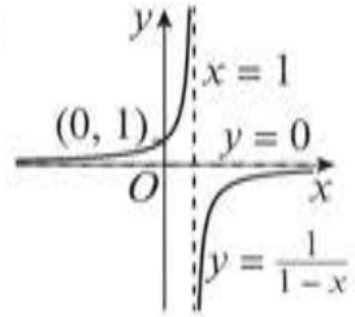
c. i


y
$y = \frac{3}{x-1}$
y = 0
O
x
x = 1
(0, −3)

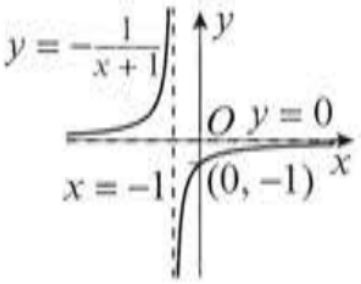
ii



iii

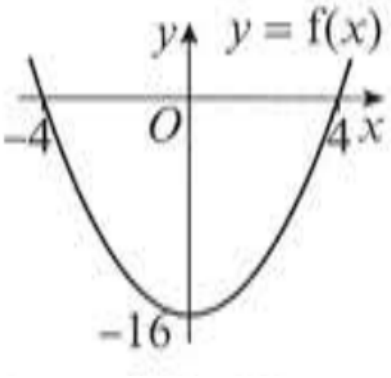


iv

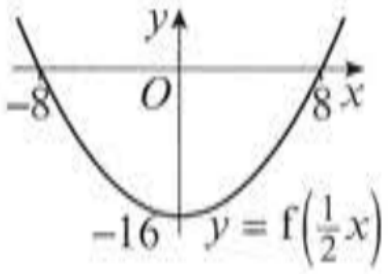


2.

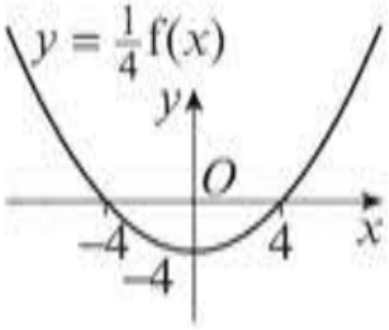
a.



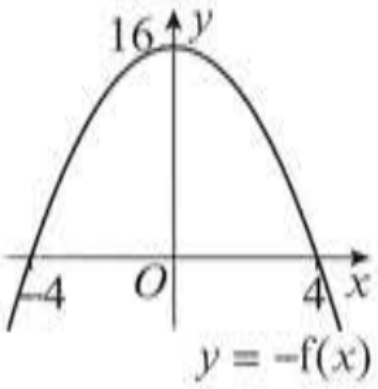
b. **i** $y = \frac{1}{4}x^2 - 16$



ii $y = \frac{1}{4}x^2 - 4$

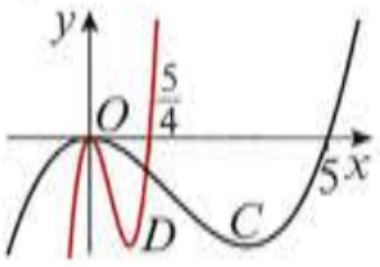


iii $y = 16 - x^2$



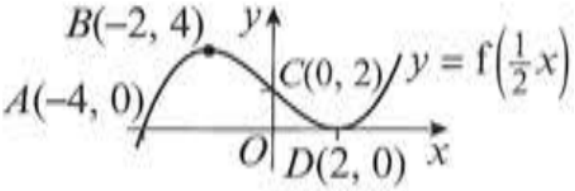
c. **i** $(0, -16)$ **ii** $(0, -4)$ **iii** $(0, 16)$

3. a, b



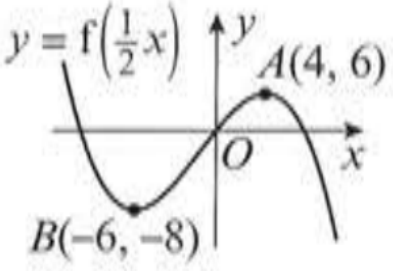
c Stretch by scale factor $\frac{1}{4}$ in the horizontal direction

4.

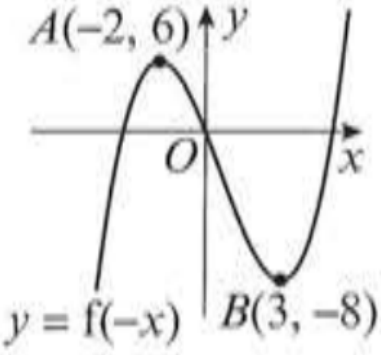

B(−2, 4)
y
C(0, 2)
y = f(½x)
A(−4, 0)
O
D(2, 0)
x

5.

a.

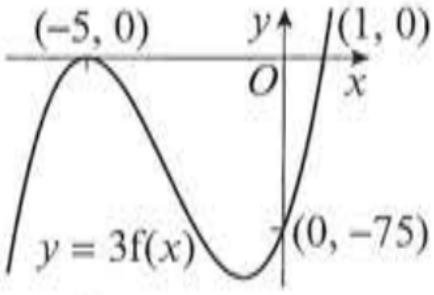

y = f(½x)
y
A(4, 6)
O
x
B(−6, −8)

b.


A(−2, 6)
y
O
x
y = f(−x)
B(3, −8)

6.

a.


(−5, 0)
y
(1, 0)
O
x
y = 3f(x)
(0, −75)

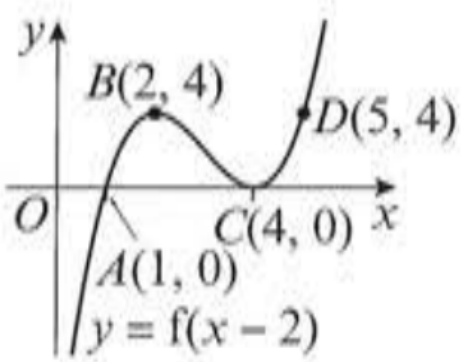
b. $y = 3(x+5)^2(x-1)$

7. **a** $a = 8$ **b** $b = \frac{1}{2}$

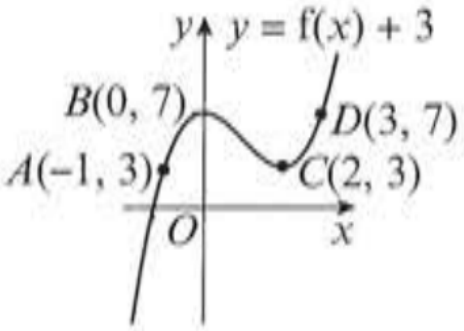
4.7 Transforming functions

1.

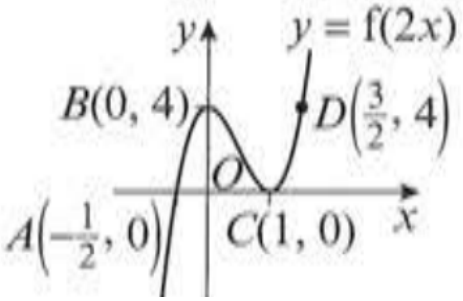
a.



b.

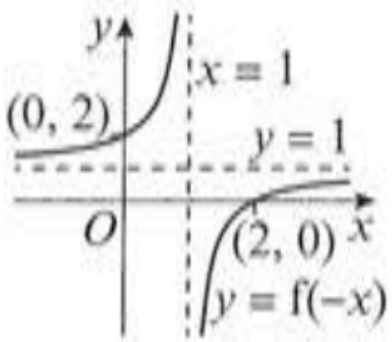


c.

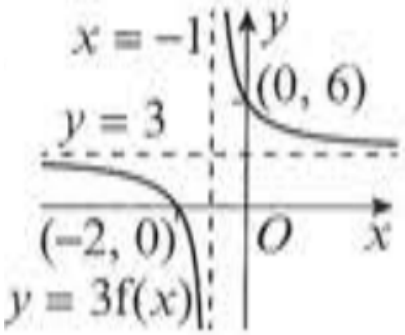


2.

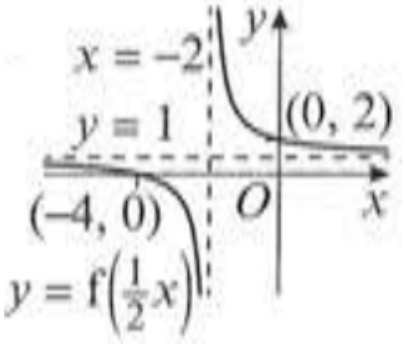
a.



b.



c.



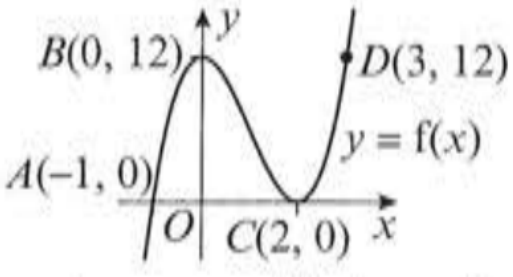
3.

a. $A(-1, 0)$, $B(0, -4.5)$, $C(1, -8)$, $D(3, 0)$

b. $A(-1, 0)$, $B(0, -36)$, $C(1, -64)$, $D(3, 0)$

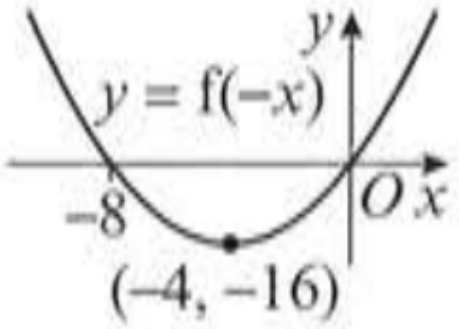
c. $A(-1, -1)$, $B(0, -10)$, $C(1, -17)$, $D(3, -1)$

4.

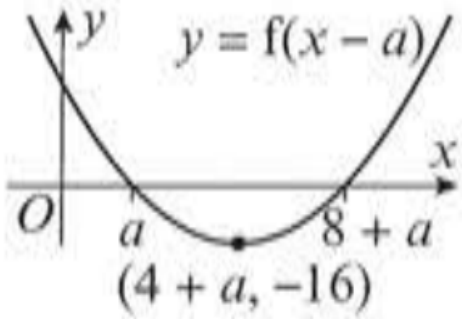


5.

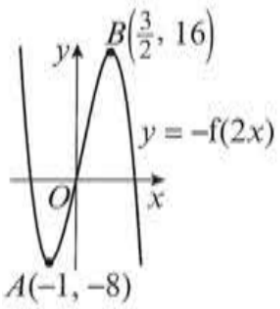
a.



b.



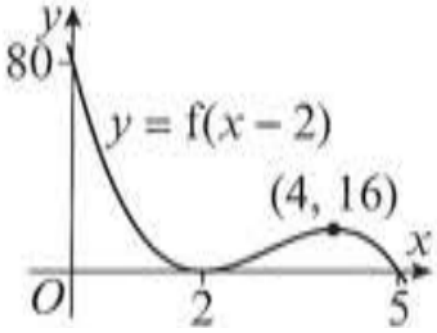
6.



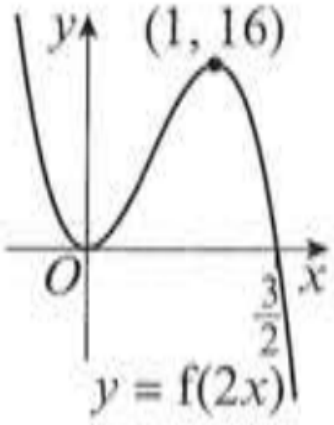
7.

a. $(3, 0)$

b. i



ii



c. -11

8. $\{a : a = -6\} \cup \{a : a \geqslant -2\}$

Problem solving: Set A

Bronze

a $b = -2$, $c = -5$, $d = 6$ **b** $(0, 6)$

Silver $b = -\frac{13}{2}$, $c = \frac{13}{2}$, $d = 5$

Gold

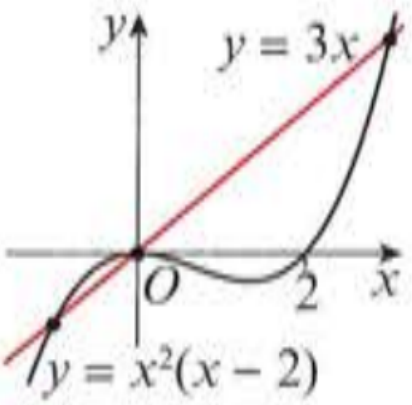
a $m = -8$, $n = 20$, $p = -16$

b $a = -2$; $y = -2x^4 + 16x^3 - 40x^2 + 32x$

Problem solving: Set B

Bronze

a



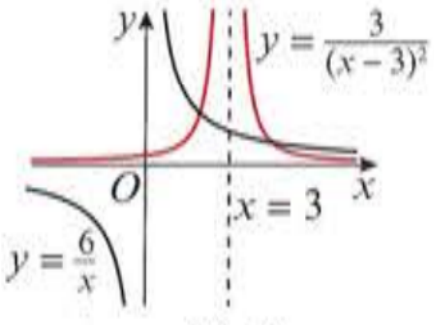
b 3

c $x^2(x - 2) = 3x \Rightarrow x^3 - 2x^2 - 3x = 0 \Rightarrow x(x + 1)(x - 3) = 0$

d $(-1, -3)$, $(0, 0)$, $(3, 9)$

Silver

a



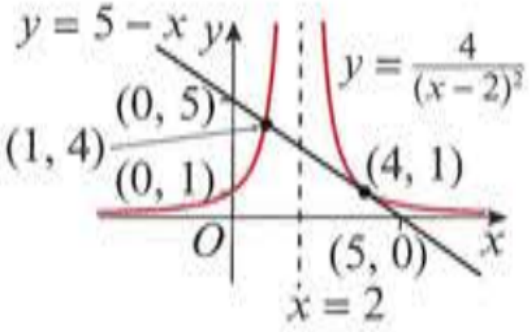
b 2

c $(2, 3)$ and $(\frac{9}{2}, \frac{4}{3})$

Gold

a $(4, 1)$

b



Selected exam question bank answers

Question 12

a. **i** 18 **ii** $c = \frac{3}{2}$

b. $f(x) = 4x^3 - 4x^2 - 15x + 18 \Rightarrow f'(x) = 12x^2 - 8x - 15$

Question 22

a. $x = \frac{3}{2}$, $x \leqslant -1$

b. $x = \frac{3}{4}$, $x = -\frac{1}{2}$

Question 33

a. $(5, -4.5)$

b. 10

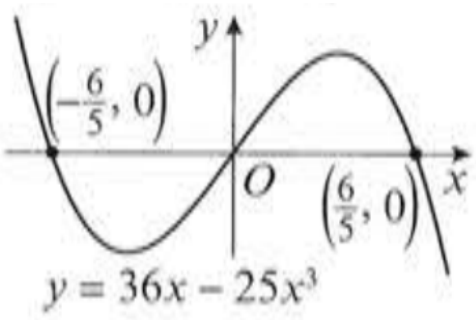
c. $x \geqslant 5$

d. $k \leqslant 1.5$

Question 49

a. $x(6+5x)(6-5x)$

b.



c. $A(-1,-11)$ and $B(1,11)$
Distance is $\sqrt{(-1-1)^2+(-11-11)^2} = 2\sqrt{122}$

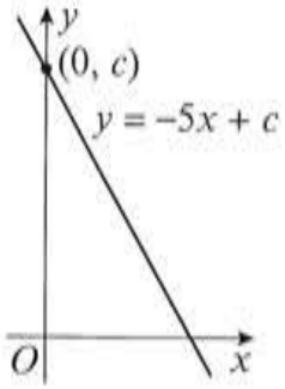
Question 72

a. $(-3,0)$, $(5,0)$

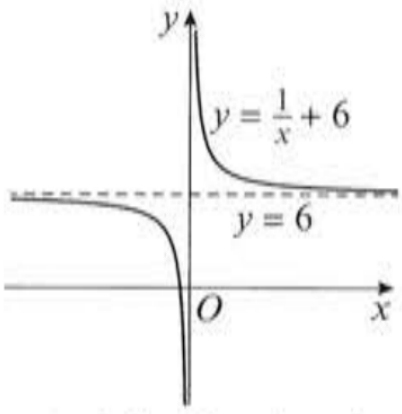
b. $\frac{863}{6}$

Question 88

a. i



ii



b. $\frac{1}{x} + 6 = -5x + c$ rearranges to $5x^2 + (6-c)x + 1 = 0$
Using $b^2 - 4ac > 0$, $(6-c)^2 - 20 > 0 \Rightarrow (6-c)^2 > 20$

c. $c < 6 - 2\sqrt{5}$ or $c > 6 + 2\sqrt{5}$